Change Your Perspective - A VR Driving Simulator to Enhance Safety Awareness in Public Road Traffic

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***Abstract*—Public road traffic is characterized by infrequent but high-risk hazards involving multiple actors, with driver distraction being a significant contributing factor. Current driver training methods, including traditional approaches and many existing VR simulators, struggle to provide safe, repeatable exposure to these critical situations or to allow learners to experience them from different viewpoints to increase awareness of other road users. This limitation hinders the development of accurate mental models of spatial relations, cross-role understanding, and anticipatory defensive strategies, while also constraining the objective evaluation of a driver's situational awareness. We present a VR driving simulator that models a real urban environment and integrates statistically motivated micro-scenarios, configurable distractions, and rule-based scoring. A key feature is a freeze-and-review mechanism that allows switching perspectives among different road users (e.g., driver, pedestrian, cyclist) when a hazard is imminent.**

This design allows for the controlled insertion of rare conflicts and distractions and provides immediate spatial inspection from alternate roles to reveal blind spots. However, the study was underpowered for detecting the impact of perspective change, and its effectiveness remains inconclusive.

The implemented VR simulator demonstrated low interaction complexity, qualitatively highlighting risks associated with blind spots and distractions. Rigorous empirical assessment of awareness gains through perspective switching is therefore a primary focus for future work.

***Index Terms*—Virtual Reality, Public Road Traffic, Training, Perspective Change, Road Safety Education, Driver Training**

I. Introduction

Participating in public road traffic is often associated with many, mostly hidden, dangers. Every road user must take care of other road users and be considerate of their own behavior. Be it the pedestrian who wants to cross the sidewalk and has to cross a road, the cyclist who has to share the sidewalk with pedestrians or the road with cars, the car driver who always has to drive at the right speed with a considerate driving style, or the truck driver who has to keep an eye on the dimensions of his truck. Each of these road users contributes to the safety of public road traffic.

The latent risk of public road traffic can occur especially in locations with concentrated vulnerable users or complex dynamics (e.g., school zones, bus stops, hospitals, and truck–cyclist interaction points). Traditional safety education relies largely on verbal instruction, static media, and limited supervised practice, which lacks: situated, repeatable exposure to hazardous but instructive edge cases, immediate performance feedback, and adaptation to differing cognitive development levels or learning styles [4], [13], [16]. Real-world practical training, while authentic, is constrained by safety, cost, logistics, and the inability to ethically reproduce genuinely dangerous scenarios [4], [10]. Immersive Virtual Reality (VR) can address these gaps by safely simulating rare, time-critical, or multi-actor conflict situations with controlled difficulty, repeatability, and instrumentation [9]. However, many existing VR road safety applications still emphasize single-role rehearsal (e.g., only the driver or only the pedestrian), limiting the development of cross-role empathy and anticipatory awareness [4]. Perspective changing (letting users seamlessly experience a scenario from different road user viewpoints (e.g., driver, pedestrian, cyclist, truck operator)) is underutilized despite its potential to (1) surface blind spots (physical and cognitive), (2) strengthen hazard mental models, and (3) encourage more considerate, defensive behavior. Consequently, there is a need for a VR training approach that (a) focuses on statistically grounded high-risk micro-scenarios, (b) integrates distraction and situational complexity, (c) supports active and observational participation, and (d) couples perspective switching with measurable behavioral feedback to evaluate awareness and mindfulness improvements. VR can serve as a platform for a digital application with which an immersive change of perspective can be realized during a simulated dangerous situation in road traffic. However, the effectiveness and learning effect of such an application are not guaranteed and require further research. The following research question (RQ) can be derived from this:

RQ: Can a VR application for simulating public road scenarios improve awareness and mindfulness among users for possible dangers with the use of perspective change?

We split our RQ into two objectives to investigate: feasibility/usability of such a system (RQ1), and measurable impact of perspective change on awareness (RQ2).

A prerequisite for the RQ is the existence of a VR application that can be used to simulate various scenarios in public road traffic. Since VR is a technology whose key characteristic is to immerse the user in the virtual world, it must be clarified how the user is going to interact with this virtual world and

be integrated. In context, it is debatable whether the user only takes on an observational role or can influence the virtual environment through their actions (and thus the outcome of a scenario). The user must therefore be able to interact with the virtual environment, which defines a second requirement. A core mechanic for the interaction is specified by the RQ in the third point. In the scenario, the user should be able to switch between the different perspectives of the road users. This defines a third requirement. In order to be able to measure the effect, there needs to be a way for the user to give feedback or measure a learning effect to determine if there was an effect. This feedback must be measurable in data so that the results can be evaluated and a provable effect can be demonstrated. This would define a fourth requirement.

The following four requirements can be summarized to formulate an answer to the RQ:

- **R1 VR application for public road scenario simulation:** The VR application should simulate many segments of a real city's public road traffic system and various traffic conditions, including peak hours and different types of vehicles, to prepare users for real-life situations.
- **R2 User's interaction and participation in the simulation:** The involvement of the user in the VR simulation can be differentiated between active and passive roles, i.e., whether the user actively controls a persona or only observes from their perspective.
- **R3 Perspective Change as Key Interaction Mechanic:** The user can switch through the perspectives of the road users and put themselves in each situation during a simulated, potentially dangerous situation.
- **R4 Feedback Measurement:** The effect of the improved awareness and mindfulness amongst users for possible dangers (in public road traffic) has to be measured in data.

The rest of the paper is structured as follows. In Section 2, we summarize and discuss related approaches within the scope of virtual driving simulation. Section 3 describes our conception and implementation, and Section 4 presents our evaluation method with results. Finally, we discuss our findings in Section 5 and conclude the paper in Section 6 with additional recommendations for future work.

II. Related Work

In this section, we first focus on general driving simulators that also worked with a focus on immersive digital twins. Building on this, we concentrate our research more on VR in this context and also look at work on VR in behavioral research for this context. This section is divided accordingly.

A. AI- and ITS-Based Driving Simulators

Research in this area integrates artificial intelligence (AI) and adaptive tutoring into immersive driving simulators.

Cheng et al. [3] review the use of driving simulators in driver education with the aim of highlighting the benefits of AI- and VR-enhanced systems. They report improved hazard perception and skill acquisition, especially when simulator immersion is high. AI techniques from digital games enable dynamic traffic scenarios and adaptive difficulty. However, they note a lack of systematic studies that focus especially on driver education and call for tighter integration of AI-driven adaptation and VR immersion.

Ropelato et al. [15] propose a VR driving simulator combined with an intelligent tutoring system. The system adapts task sequencing using performance metrics such as lane-keeping and reaction time. Results show good usability and presence, but simulator sickness led to participant dropout (25%), indicating usability and session-length limitations.

B. VR Serious Games and Educational Platforms for Road Traffic (Non-AI Focus)

These approaches concentrate on immersive learning environments and serious games.

Leon-Paredes et al. [12] introduce the MUC-UPS-EMOV platform, which combines VR-based serious games with a web-based mobility observatory. Training modules are derived from real accident data and target high-risk behaviors (e.g., drunk driving, high speeding). The platform promotes awareness and community engagement, but empirical evidence of long-term accident reduction is not yet demonstrated.

Yeo et al. [8] evaluate a VR pedestrian safety program for children. The study shows improvements in street-crossing behavior and highlights VR's ability to support personalized feedback. However, long-term retention and contextual generalization remain open questions.

Kim et al. [1] present a VR-based safety education program for young children. Teachers reported improved safety knowledge and high engagement, with immersion identified as a key factor. Evaluation relied primarily on subjective teacher feedback rather than behavioral metrics.

Rahim et al. present a "Road Safe VR" [18], a low-cost mobile VR game for pedestrian training. The system includes structured lessons and endless gameplay modes. While affordable and accessible, validation was conducted with a small and diverse sample, which limits external validity.

C. VR for Behavioral Research in Public Road Traffic

This part of the related work investigates whether VR reliably reproduces real-world behavior and supports safety research.

Bhagavathula et al. [2] compare pedestrian behavior in real and virtual environments. They find similar crossing decisions and distance estimations across conditions, supporting VR's ecological validity. However, participants overestimated vehicle speed in VR and reported lower presence than in real-world settings.

Ye et al. [19] study pedestrian behavior under unfamiliar traffic rules in VR (left- vs. right-hand traffic). Results show that habitual gaze behavior increases risk, and VR provides a safe environment for studying such effects. The study focuses on short-term behavioral observation rather than intervention outcomes.

Kwon et al. [11] analyze how urban street attributes influence perceived risk and crossing behavior. Features such as sidewalk separation reduce perceived risk, whereas poor visibility and lighting increase it. Higher perceived risk affects decision time and gait. While VR enables controlled manipulation of environmental factors, real-world transfer is not assessed.

Vankov et al. [17] provide a systematic review of 39 studies using VR in road safety research. Findings suggest improved realism and hazard perception compared to non-immersive methods. However, simulator sickness is common, and most studies assess short-term perception rather than long-term behavioral change.

III. Concept and Implementation

Our VR application places the user behind the wheel of a motor vehicle in three predefined road traffic scenarios, each featuring dangerous situations and deliberate distractions. When an accident is about to occur, the simulation freezes and allows the user to cycle through the perspectives of every involved party (driver, pedestrian, or cyclist) so they can understand the full picture before the scenario resumes. If the user avoids the hazard, the scenario simply ends. This design addresses our four requirements: realistic everyday traffic situations (R1), active vehicle control combined with perspective switching (R2), perspective change as the central awareness mechanism (R3), and a penalty-based feedback loop for measurable outcomes (R4).

A. Simulation Environment

The simulation recreates a realistic urban traffic setting. Users wear a VR headset (Meta Quest 2 or 3) to see a stereoscopic 3D view of the virtual world, use handheld controllers to accelerate and brake, and hear spatial audio engine sounds, ambient traffic, and passenger speech that reinforces the sense of being on a real road. Within this environment, autonomous traffic participants (other vehicles, pedestrians, and cyclists) follow behavior patterns that mimic everyday traffic flow. The system continuously monitors the user's driving behavior and detects traffic violations or collisions in real time. When a dangerous situation unfolds, the scene can freeze automatically, allowing the user to switch between the viewpoints of all involved road users before the scenario resumes. A penalty-point system provides immediate on-screen feedback, giving users a clear and quantifiable measure of their performance. To keep interaction accessible, controls are intentionally minimal: accelerate, brake, switch perspective, change radio station, and answer the phone. Because the vehicle follows a fixed route, audiovisual turn signals indicate upcoming turns, which also helps reduce motion sickness. Functioning rear-view and side mirrors support spatial orientation. A second person at the PC can manage scenario selection and settings, including an optional 18+ toggle for more graphic accident visuals. A dedicated training scenario teaches all controls before the real sessions begin.

B. Scenario Design

1) Rationale for Scenario Selection: We reviewed European (CARE, 2022) [5], U.S. Fatality Analysis Reporting System (FARS) (2013–2017) [7], and pedestrian injuries due to mobile phone usage in public places [14]. The key findings were that pedestrian fatalities cluster around road crossings and bus stop contexts; drivers feel most nervous in school zones and at intersections; right-turn conflicts between trucks and cyclists remain a severe blind-spot problem; and common distractions like phone use and radio adjustment keep appearing as risky behaviors. To address these high-frequency conflict types, cover different VRU categories (pedestrians, cyclists), and exercise varied cognitive demands within one training session, we distilled three complementary scenarios: (1) School Zone, (2) Intersection, and (3) Blind Spot. Every scenario shares a common structure—the user drives with a chatty passenger who acts as a constant distraction.

- **School Zone Scenario (SZS):** The user drives through a traffic-calmed school zone and merges onto a main road. Two hazards arise: a bus pulling in front of them toward a bus stop and a pedestrian dashing across without looking to catch that bus. A cell phone in the car starts ringing right before the crossing. Possible mistakes: picking up the phone, hitting a pedestrian, or colliding with the bus by ignoring its right of way.
- **Intersection Scenario (IS):** The user approaches an intersection to turn right, but another car speeds up and cuts them off. A loud radio—whose signal drops out just before the turn—serves as the distraction. Possible mistakes: failing to drive defensively and crashing or fiddling with the radio at the critical moment.
- **Blind Spot Scenario (BSS):** Here, the user drives a truck approaching an intersection while a cyclist rides alongside on the right. As the truck turns right and the cyclist goes straight, the cyclist ends up in the truck's blind spot. If the user neglects to check, an accident occurs.

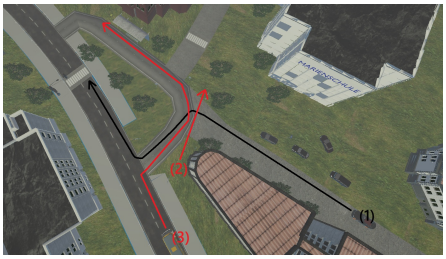
A user study was chosen to validate the research question; its design is described alongside the evaluation.

C. Scoring System

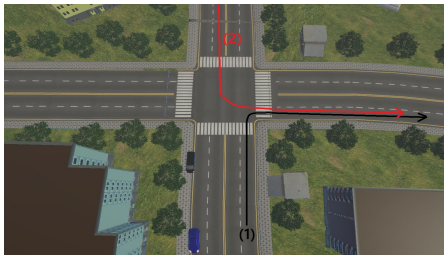
For R4, we built a penalty-point system modeled on the German driver-license theory test [6]. Points scale with offence severity: right-of-way violations earn 4–5 points, phone use 4, radio use 3, and blind-spot failures 4. The resulting totals per scenario are SZS: 13 (right-of-way 4 + pedestrian hit 5 + phone 4), IS: 7 (defensive driving 4 + radio 3), and BSS: 4 (blind spot 4).

D. Technical Realisation

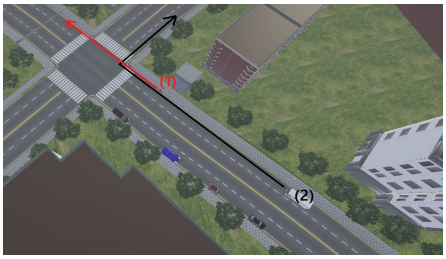
The application was built in Unity with 3D environments modeled in Blender to faithfully recreate the street layout, including buildings, sidewalks, and traffic infrastructure. It runs on Meta Quest 2 and 3 headsets with standard touch controllers on a Windows PC. The system is organized into six functional layers:



(a) School Zone: (1) user vehicle, (2) pedestrian, (3) bus



(b) Intersection: (1) user vehicle, (2) cutting-in car



(c) Blind Spot: (1) truck, (2) cyclist in blind spot

Fig. 1: Concept top-down views of the three scenarios.

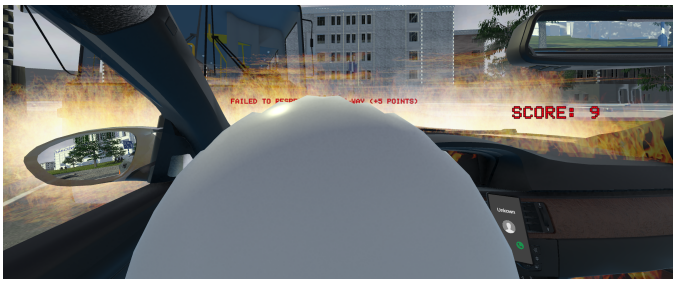


Fig. 2: Vehicle collision with deployed airbag and damage effects.

The **hardware and platform layers** provide the physical interface. The VR headset tracks the user's head position and orientation in six degrees of freedom, while the controllers capture inputs such as trigger presses for acceleration and braking. The operating system and VR runtime translate these raw sensor signals into standardized tracking data for the rendering engine.

The **engine layer** handles real-time rendering of the 3D environment, physics simulation for vehicle motion and collisions, spatial audio for engine sounds and environmental cues, as well as user interface elements such as the speedometer, turn indicators, and penalty notifications.

The **simulation layer** governs the behavior of all traffic participants. Vehicles follow predefined curved paths with adjustable speed, while pedestrians navigate waypoint networks with randomized lateral offsets for natural variation. The collision detection system identifies near misses, right-of-way breaches, and impacts, forwarding these events to the scoring and freeze mechanisms. When a collision occurs, the vehicle stops realistically—airbags deploy, glass shatters, and damage effects appear—providing visceral feedback on the consequences of distracted or careless driving (Figure 2).

The **application layer** handles user-facing flow: a main menu for scenario selection and settings (including the 18+ toggle), a training scenario that progressively introduces controls, and the three main scenarios with their respective configurations.

The **core management layer** orchestrates cross-scenario services. The perspective manager switches the camera between different road users' viewpoints at the press of a button, enabling the user to experience, for example, a pedestrian's perspective after nearly being struck. The score manager tracks infractions, displays penalty notifications with automatic fade-out, and presents a summary at the end of each scenario. The distraction system triggers phone calls and radio interruptions at scripted moments to challenge the user's attention, creating the cognitive load that makes real-world driving dangerous. In summary, the overall pipeline proceeds as follows: hazard detection (simulation layer) $\rightarrow$ scene freeze, $\rightarrow$ perspective switch (core management layer) $\rightarrow$ scoring and on-screen feedback.

IV. Evaluation

To address our research question—"Can a VR application for simulating public road scenarios improve awareness and mindfulness amongst users for possible dangers with the use of perspective change?"—we defined four requirements: R1 (realistic traffic simulation), R2 (active vs. passive user roles), R3 (perspective change as core mechanic), and R4 (measurable feedback on awareness). From these, we derived three hypotheses:

- **H1** (Realism and presence): The application creates a convincing, immersive road environment.
- **H2** (Usability of interaction): Simple controls make the application intuitive and accessible.
- **H3** (Impact of perspective change): Switching viewpoints helps users understand dangerous situations more deeply, triggered by the feedback mechanism.

H1 maps to R1, H2 to R2, and H3 to R3 and R4.

A. User Study Design

We recruited 24 participants (n=24) for the user study; each participant goes through all three implemented scenarios. To get insights about the effectiveness of the perspective change, the participants have been split into two separate groups. The first group experienced the scenarios with the perspective change, and the second group without the perspective change. The order of scenarios has been randomized for each participant to ensure no bias from the scenario order. The study process is divided into several steps that each participant had to go through. First, each participant must complete the persona

and pre-study questionnaire. Afterwards, each participant will receive an introduction to the VR application, supported by the training scenario, so that they can learn how to control it and become familiar with the VR application. Then they moved straight on to the first selected scenario. After the participant has completed the scenario, their first impressions are recorded by completing the NASA-TLX, SUS, and IPQ questionnaires. Once the questionnaires have been completed, each participant goes through the remaining two selected scenarios. Finally, the participant is asked the questions from the post-study questionnaire in the form of an interview. The interview is seen here as an opportunity to engage in conversation with the participant in order to understand the impression left, gather ideas for improvements, and collect constructive criticism.

B. Results

1) Demographics and Pre-Study Behavior: We recruited 24 participants, mostly aged 19–25 (only two under 18). Daily road users made up 62.5%, and 91.7% held a driving license. VR experience varied widely (54% with some exposure, 25% none), and motion sickness susceptibility ranged from "not at all" (42%) to "severely" (4%). Pre-study questionnaires indicated generally defensive, considerative driving habits, though compliance with speed limits and phone usage was mixed. Most participants mainly noticed others breaking rules rather than themselves. Baseline ratings for empathy, perceived risk, and the value of VR safety training averaged 3–4 out of 5, providing a solid baseline for evaluation.

2) NASA Task Load Index (TLX) Results: Figure 3 summarizes cognitive and physical workload. Every dimension stayed in the low range (0–6.7). Performance concern scored highest (6.6), suggesting genuine engagement rather than passive observation (R2). Mental demand came next (3.9), consistent with the cognitive effort of processing multiple perspectives (R3). Physical demand was lowest (2.1), confirming an accessible interface (R1). Temporal demand (3.2), effort (3.8), and frustration (3.4) all stayed low, meaning participants could focus on learning instead of wrestling with usability (R4). The ANOVA was significant (F=5.34, p=0.0002), validating that performance concern stood out as the dominant workload factor—a good sign for perspective-driven awareness.

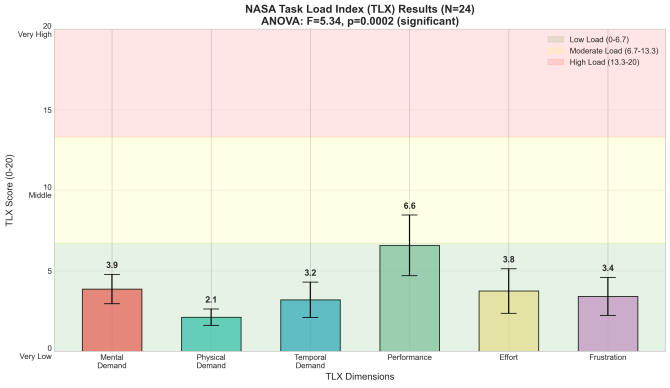


Fig. 3: NASA Task Load Index (TLX) bar chart.

3) System Usability Scale (SUS) and Igroup Presence Questionnaire (IPQ) Results: The SUS score was 73.5/100, indicating good usability and confirming that interaction issues did not bottleneck the awareness-building purpose (H2). The IPQ yielded a moderate overall presence score of 3.32 out of 5, indicating sufficient immersion for training (H1, R3). Participants maintained a reasonable balance between immersion and real-world grounding (R1). The highest-rated dimensions were "Virtual World Realism" and "Sense of Being There" (both 3.46), while "Attention to Real Environment" scored lowest (2.92), favorably indicating users were absorbed in the virtual setting.

4) Post-Study Results: Post-study data showed the VR experience actively improved or maintained safety awareness, with average ratings for empathy and awareness at 3.6/5 ($\pm$1.00), change in risk perception at 3.4/5 ($\pm$0.53), and overall experience at 4.0/5 ($\pm$0.24), corresponding to "Good." Every participant (100%) recommended the system. The School Zone Scenario (SZS) was considered the most distracting (42%), followed by the Intersection (29%) and Blind Spot (17%). In interviews, participants highlighted the SZS as the most memorable due to its rapid event sequence and numerous distractions. Positive feedback praised the realistic driver's view, intuitive controls, natural driving feel, and well-integrated accident mechanics. The virtual environment—including street layouts and pedestrians—was perceived as highly believable. However, study leaders noted that participants often drove cautiously below the speed limit, which frequently prevented pre-programmed hazards from triggering. Consequently, common participant suggestions included making scenarios longer, allowing manual steering, adding heavier ambient traffic, and chaining scenarios into a single extended drive to exploit dwindling attention over time.

5) Learning Effect and Perspective Change: Average penalty points per scenario were low (SZS: 1.5 ± 1.0, IS: 0.5 ± 0.5, BSS: 1.0 ± 0.7). While penalties descriptively declined by 38.5% across ordered sessions (from 1.1 down to 0.7), neither this learning effect (p=0.4109) nor differences between scenario types (p=0.2158) achieved statistical significance. This reflects insufficient statistical power caused by short scenario durations ($\approx$1 min) and overly cautious driving, leading to a zero-inflated penalty distribution that restricted measurable improvement. Consequently, among the 12 participants assigned to the perspective-change condition, very few actually triggered the freeze-and-review mechanic. The study is thus underpowered to evaluate the impact of perspective change (H3); reliably detecting a medium effect (d=0.50) would require a significantly larger sample size ($n \approx 34$ minimum).

V. Discussion

The resulting VR application demonstrated high user-friendliness and accessibility. The SUS score (73.5) and low task load (NASA-TLX) verify our hypotheses regarding usability and realistic participation (H2). Furthermore, the moderate IPQ score (3.3/5) confirms that the application

creates a sufficiently convincing sense of presence (H1). The integrated distraction mechanisms functioned as intended, with the School Zone Scenario rated as the most distracting. Together, these findings support our first research objective: establishing the feasibility and usability of the system. Regarding the impact of perspective change on safety behavior (H3), statistical differences between scenarios were not established (ANOVA p=0.216), and learning over time proved non-significant (p=0.411). This lack of statistical power stemmed primarily from a low event rate and numerous zero-penalty scores. Participants exhibited overly cautious driving behavior—despite admitting to real-world traffic violations—likely due to the novelty of the VR environment and a slow acclimatization process. The short scenario length (approx. 1 minute) meant participants retained focus on basic controls rather than succumbing to the cognitive load of distractions. Consequently, our data is insufficient to definitively support H3 or the second objective of our RQ. Nevertheless, aligning with prior studies [1], [17]–[19], our findings indicate that VR is a viable medium for simulating dangerous traffic situations. The beginner-friendly design suggests promising applications for learner drivers and schoolchildren. For perspective change mechanisms to be effectively evaluated in the future, applications must employ longer scenarios to overcome initial acclimatization effects and naturally provoke varied driving behavior.

A. Threats to Validity

Several factors limit our findings. First, the short scenario duration combined with a slow VR acclimatization process induced artificially cautious behavior, separating virtual performance from real-world driver habits. This caused a heavy floor effect in penalty points, critically limiting the statistical power to evaluate H3. Second, the between-groups design allocated only 12 participants per group, which is insufficiently powered to detect small-to-medium behavioral effects; future work requires a within-subjects design or a substantially larger sample. Third, our evaluation relies on short-term data, leaving the real-world transfer of behavior unmeasured. Fourth, the demographically narrow participant pool affects broader generalizability. Finally, repeatedly administering the NASA-TLX, SUS, and IPQ questionnaires after every scenario may have induced respondent fatigue, which could be mitigated by using aggregated post-session measures.

VI. Conclusion & Future Work

This work introduced a modular VR driving simulator with three statistically grounded micro-scenarios and a freeze-and-review perspective change mechanism to build cross-role awareness. A study with 24 participants demonstrated good usability (SUS 73.5/100), low workload (NASA-TLX), moderate presence, and a 100% recommendation rate, confirming the system's feasibility and high user acceptance. However, the behavioral effect of perspective change remained inconclusive, primarily due to short scenario durations, low event rates, and cautious driving that yielded zero-inflated penalty counts. Consequently, our main contribution lies in the proven system architecture and evaluation protocol rather than definitive proof of perspective-change effectiveness. To rigorously evaluate the core mechanic, future research must increase event density through longer, adaptively harder scenarios and pair the freeze-and-review feature with playback mechanisms.

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